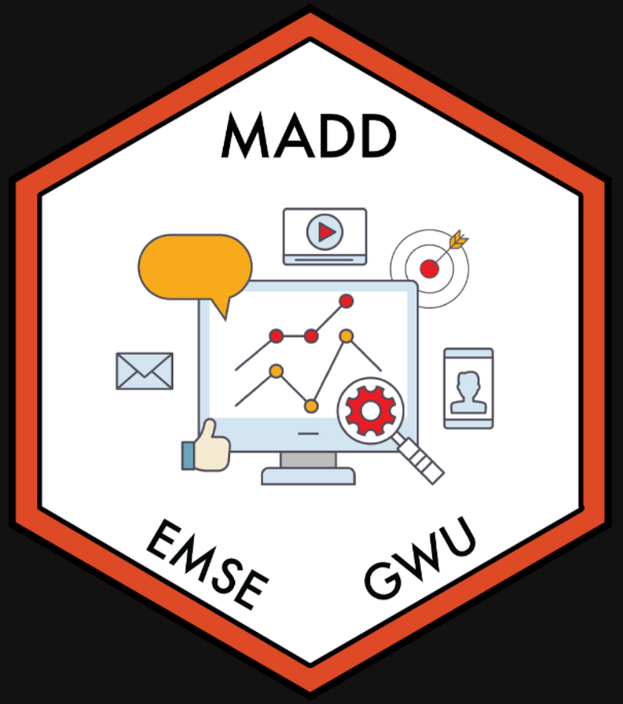


DOE & Power Analysis

EMSE 6035: Marketing Analytics for Design Decisions

John Paul Helveston

November 04, 2026

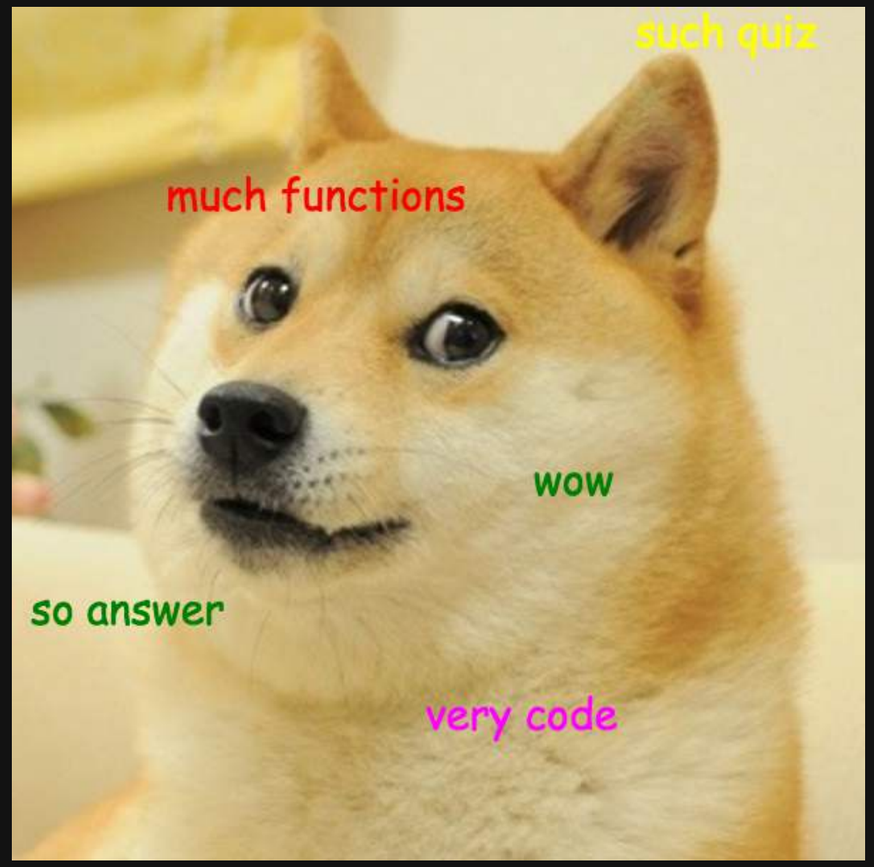
Quiz 4

10:00

Write your name on the quiz!

Rules:

- Work alone; no outside help of any kind is allowed.
- No calculators, no notes, no books, no computers, no phones.



Week 11: *DOE & Power Analysis*

1. Design of Experiment

2. Design Efficiency

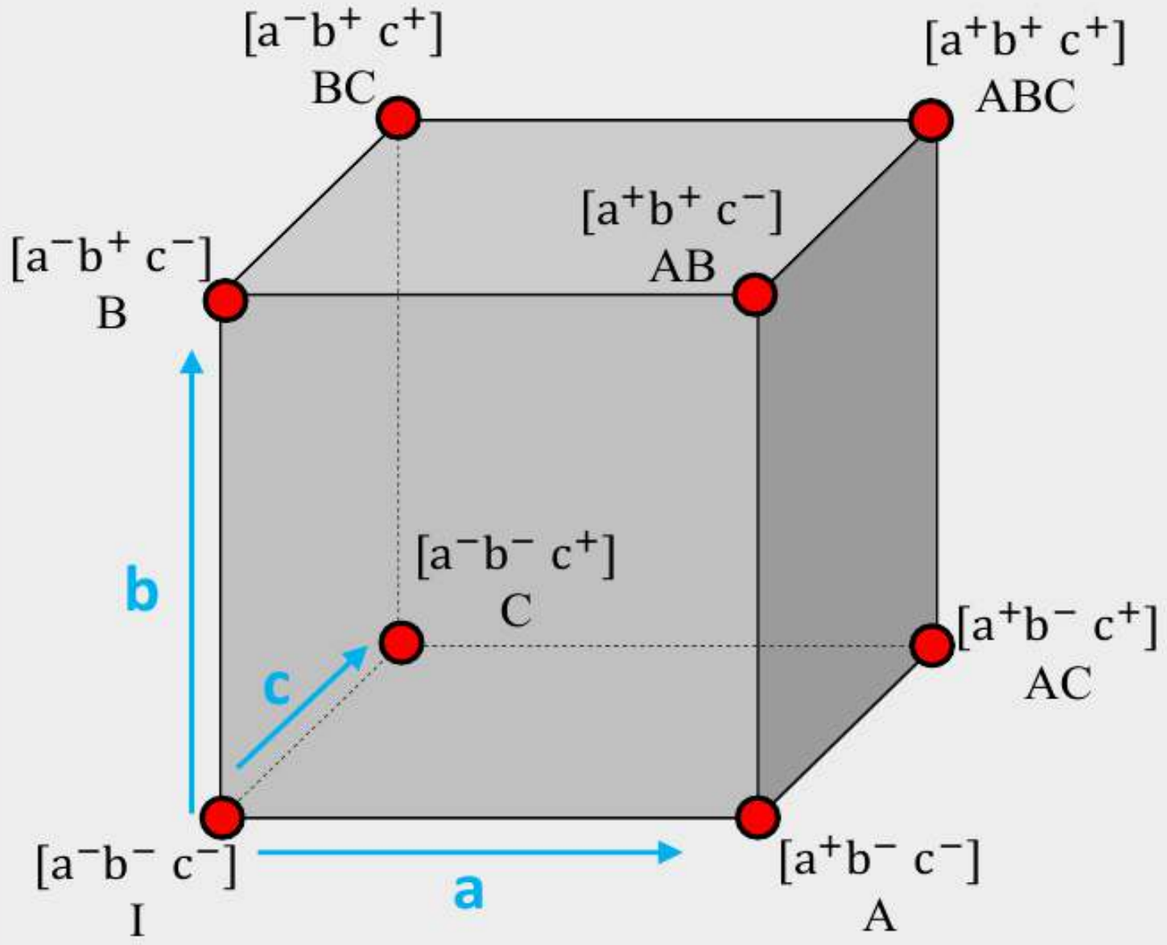
3. Power Analysis

Week 11: *DOE & Power Analysis*

1. Design of Experiment
2. Design Efficiency
3. Power Analysis

Main & Interaction Effects

Full design space for 3 effects: A, B, C



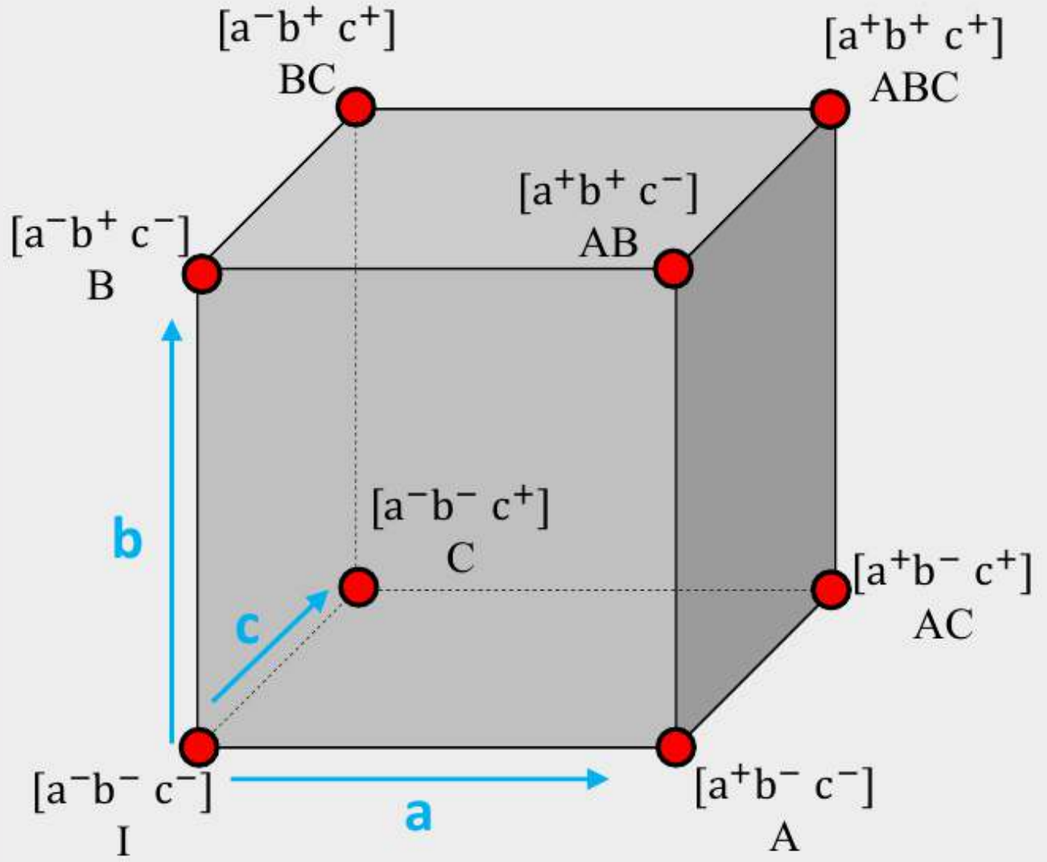
Full design space for 3 effects: A, B, C

Example: *Cars*

A: Electric? (Yes+ or No-)

B: Warranty? (Yes+ or No-)

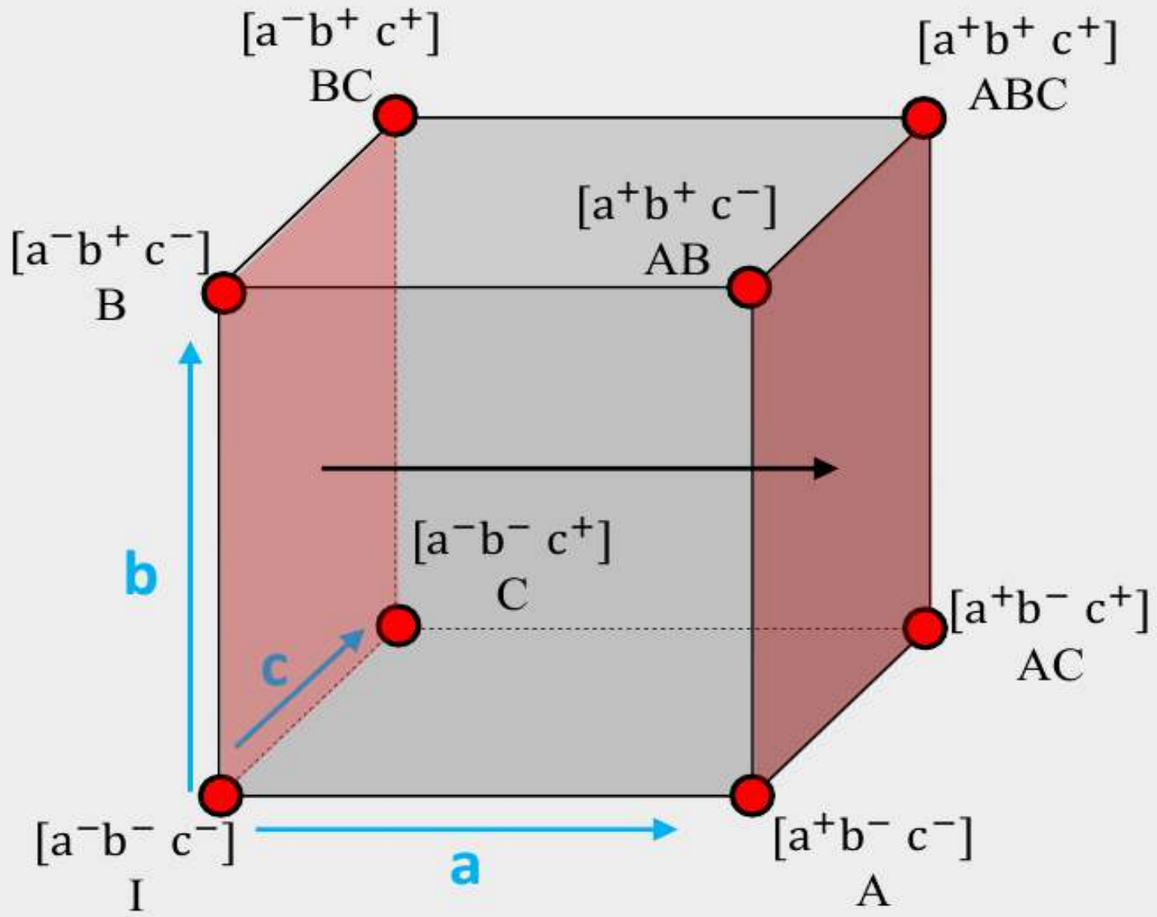
C: Ford? (Yes+ or No-)



Main Effects

$$ME(a) = \left(\frac{A + AB + AC + ABC}{4}\right) - \left(\frac{I + B + C + BC}{4}\right)$$

(A: Electric? Yes+ or No-)

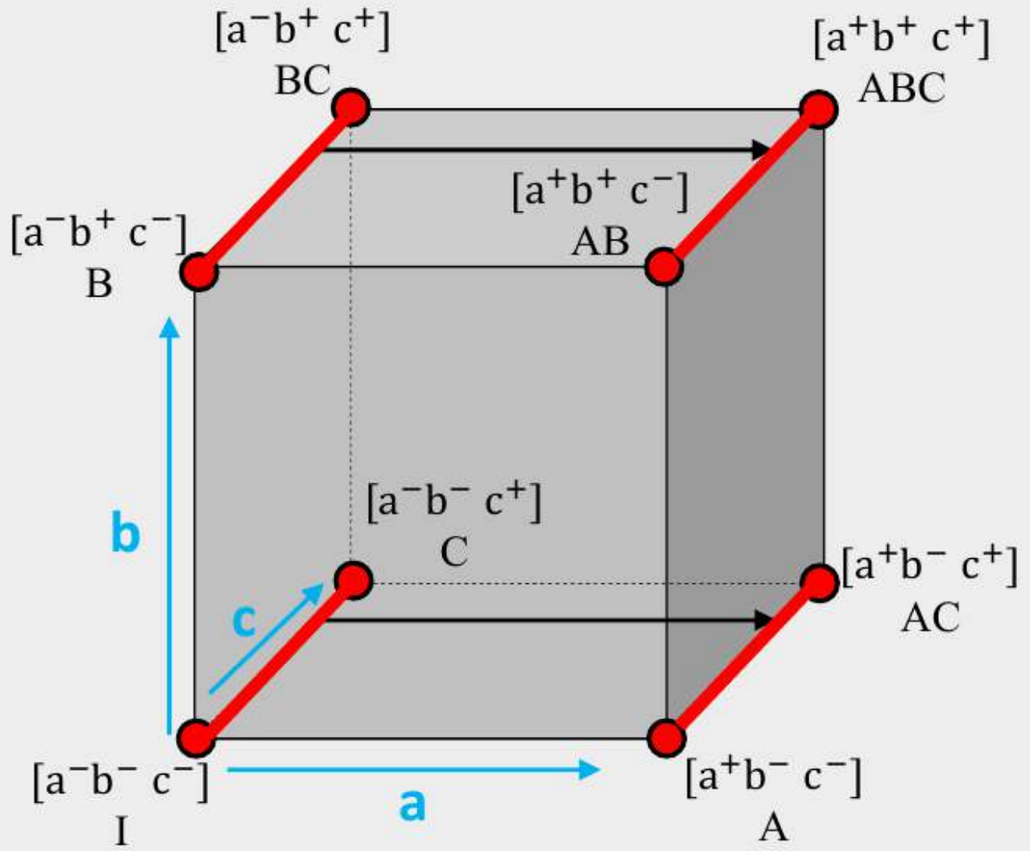


Interaction Effects

$$INT(ab) =$$

$$\frac{1}{2}\left[\left(\frac{AB + ABC}{2}\right) - \left(\frac{B + BC}{2}\right)\right] -$$

$$\frac{1}{2}\left[\left(\frac{A + AC}{2}\right) - \left(\frac{I + C}{2}\right)\right]$$



Example: Wine Pairings

meat	wine
fish	white
fish	red
steak	white
steak	red

Main Effects

1. `meat`: **Fish** or **Steak**?
2. `wine`: **Red** or **White**?

Example: Wine Pairings

meat	wine
fish	white
fish	red
steak	white
steak	red

Main Effects

1. `meat`: **Fish** or **Steak**?
2. `wine`: **Red** or **White**?

Interaction Effects

1. `meat*wine`: **Red** or **White** wine *with **Steak***?
2. `meat*wine`: **Red** or **White** wine *with **Fish***?

Open `interactions.qmd`

Fractional vs Full Factorial Designs

Full Factorial Design

Example: *Cars*

A: Electric? (Yes+ or No-)

B: Warranty? (Yes+ or No-)

C: Ford? (Yes+ or No-)

```r
library(cbcTools)

profiles <- cbc_profiles(
    electric = c(1, 0),
    warranty = c(1, 0),
    ford     = c(1, 0)
)
```

```r
profiles
```

```
#>   profileID electric warranty ford
#> 1         1        1        1    1
#> 2         2        0        1    1
#> 3         3        1        0    1
#> 4         4        0        0    1
#> 5         5        1        1    0
#> 6         6        0        1    0
#> 7         7        1        0    0
#> 8         8        0        0    0
```

Full Factorial Design

Balanced?

All levels appear an equal number of times.

Orthogonal?

All pairs of levels appear together an equal number of times.

```r
library(cbcTools)

profiles <- cbc_profiles(
    electric = c(1, 0),
    warranty = c(1, 0),
    ford     = c(1, 0)
)
```

```r
profiles
```

```
#>   profileID electric warranty ford
#> 1         1        1        1    1
#> 2         2        0        1    1
#> 3         3        1        0    1
#> 4         4        0        0    1
#> 5         5        1        1    0
#> 6         6        0        1    0
#> 7         7        1        0    0
#> 8         8        0        0    0
```

Fractional Factorial Design

Balanced?

All levels appear an equal number of times.

Orthogonal?

All pairs of levels appear together an equal number of times.

```
1 profiles[c(1, 3, 5, 6),]
```

```
#>   profileID electric warranty ford
#> 1         1        1        1    1
#> 3         3        1        0    1
#> 5         5        1        1    0
#> 6         6        0        1    0
```

Comparing Full and Fractional Factorial Designs

Open `balance-orthogonality.qmd`

Practice Question 1

Consider the following experiment design

a	b	c	Effect
+	-	-	A
-	+	-	B
+	-	+	AC
-	+	+	BC

a. Is the design balanced? Is is orthogonal?
b. Write out the equation to compute the main effect for a, b, and c.
c. Are any main effects confounded? If so, what are they confounded with?

Week 11: *DOE & Power Analysis*

1. Design of Experiment
2. Design Efficiency
3. Power Analysis

A simple conjoint experiment about *cars*

Attribute	Levels
Brand	GM, BMW, Ferrari
Price	$20k, $40k, $100k

Design: 9 choice sets, 3 alternatives each

```
#> Attribute counts:
#>
#> brand:
#>
#>      GM     BMW Ferrari
#>      10      11       6
#>
#> price:
#>
#>  20k  40k 100k
#>    8   10    9
```

```
#> Pairwise attribute counts:
#>
#> brand & price:
#>
#>             20k 40k 100k
#>   GM          3   5    2
#>   BMW         4   4    3
#>   Ferrari     1   1    4
```

A simple conjoint experiment about *cars*

Attribute	Levels
Brand	GM, BMW, Ferrari
Price	$20k, $40k, $100k

Design: 90 choice sets, 3 alternatives each

```
#> Attribute counts:
#>
#> brand:
#>
#>      GM     BMW Ferrari
#>      93      89      88
#>
#> price:
#>
#>  20k  40k 100k
#>   93   91   86
```

```
#> Pairwise attribute counts:
#>
#> brand & price:
#>
#>            20k 40k 100k
#>   GM        39  27   27
#>   BMW       33  31   25
#>   Ferrari   21  33   34
```

Bayesian D-efficient designs

Maximize information on “Main Effects” according to priors

Attribute	Levels	Prior
Brand	GM, BMW, Ferrari	0, 1, 2
Price	$20k, $40k, $100k	0, -1, -4

$$v_j = 1\delta^{\text{BMW}} + 2\delta^{\text{Ferrari}} - 1\delta^{\text{40k}} - 4\delta^{\text{100k}}$$

Bayesian D-efficient designs

Maximize information on “Main Effects” according to priors

Attribute	Levels	Prior
Brand	GM, BMW, Ferrari	0, 1, 2
Price	$20k, $40k, $100k	0, -1, -4

```
#> Attribute counts:
#>
#> brand:
#>
#>      GM     BMW Ferrari
#>     100      90      80
#>
#> price:
#>
#>  20k  40k 100k
#>   80  120   70
```

```
#> Pairwise attribute counts:
#>
#> brand & price:
#>
#>             20k 40k 100k
#>   GM         40  50   10
#>   BMW        30  40   20
#>   Ferrari    10  30   40
```

Negative of the hessian evaluated at a set of parameters is called the **“Information Matrix”**

$$\boldsymbol{I}(\boldsymbol{\beta}) = -\nabla^2_{\boldsymbol{\beta}} \ln(\square)$$

“D-optimal” designs attempt to minimize the “D-error” of a design

$$D = |\boldsymbol{I}(\boldsymbol{\beta})|^{-1/p}$$

where p is the number of coefficients in the model

Getting the D-error of a design

`cbc_inspect()` and `cbc_compare()` report a design's D-error

But only for designs built with a D-optimal method (`"stochastic"`, `"modfed"`, `"cea"`) - D-error gets computed *during* the optimization

(random designs have no D-error to report)

Finding Efficient Designs

Open `design-efficiency.qmd`

Your Turn

15:00

1. Individually, create a Bayesian D-efficient fractional factorial survey design. Inspect the attribute balance and overlap.
2. Try all three D-optimal methods and compare them with `cbc_compare()`. Which gives the lowest D-error? Which is the most balanced?
3. Compare your results with your teammates.

Week 11: *DOE & Power Analysis*

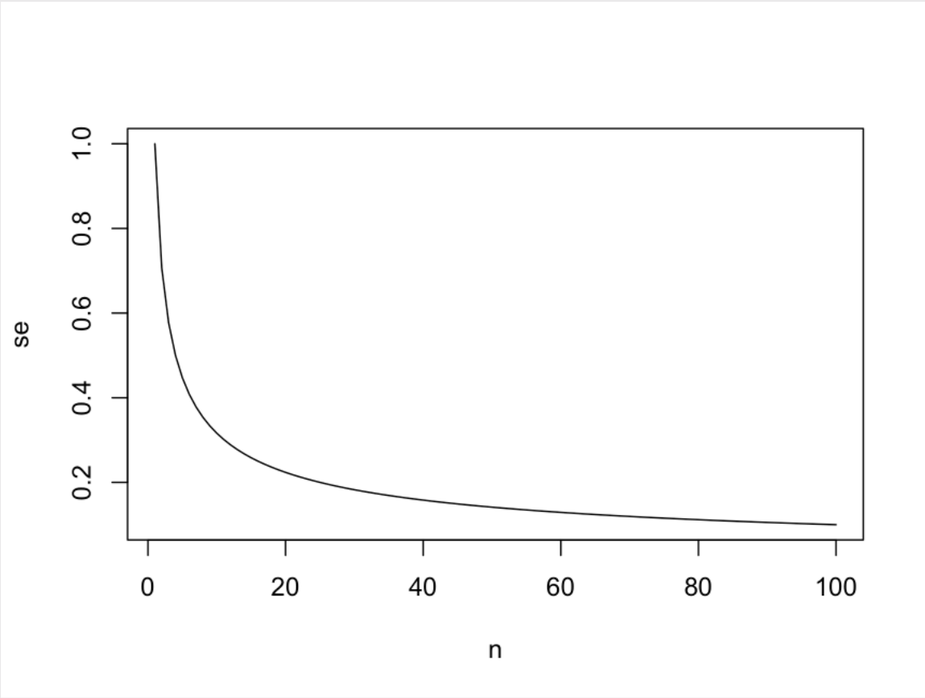
1. Design of Experiment
2. Design Efficiency
3. Power Analysis

How many respondents do I need?

How many respondents do I need
to get X level of precision on β?

Standard errors are inversely related to $\sqrt{N}$

```
1 n <- seq(100)
2 se <- 1/sqrt(n)
3 plot(n, se, type = "l")
```



Standard errors also decrease with:

- Fewer attributes
- Fewer levels in each categorical attribute
- More questions per respondent

Using {cbcTools}, we can run simulations to determine the necessary sample size for a specific model

Open `power-analysis.qmd`

Your Turn

15:00

Individually:

1. Using the survey design you created in the last practice, conduct a power analysis. Use `summary()` to find the sample size you need to reach 80% power (at $\alpha = 0.05$) on every parameter.
2. Re-run it with a more conservative set of priors. How much does your required sample size change?
3. Compare your results with your teammates.